THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 1–4: **The Wonderful World of Science, Diversity in the Living World, Mindful Eating: A Path to a Healthy Body, Exploring Magnets** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Which combination in a vegetarian diet gives good-quality protein? 1

(a) water with salt **(b)** sugar with butter
(c) cereals with pulses **(d)** only fruits

2. Why does a steel almirah become slightly magnetic over many years? 1

(a) it is kept dry **(b)** it is painted
(c) it is exposed to sunlight **(d)** Earth's magnetism slowly magnetises it

3. Which one of these questions leads to a hypothesis most directly? 1

(a) Why do wet clothes dry faster on a windy day?
(b) What is the date today?
(c) Who is taller?
(d) What do you want to eat?

4. Which of these shows interdependence between plants and animals? 1

(a) plants eat animals **(b)** animals use oxygen produced by plants
(c) animals make plants **(d)** plants hunt

5. Which type of food would best help recovery after a long illness? 1

(a) only sweets **(b)** spicy fried food
(c) easily digestible, nutritious food **(d)** no food

6. If the Earth's geographic north pole region behaves like a magnetic south pole, a compass's north needle is attracted to it because: 1

(a) unlike poles attract **(b)** wind
(c) gravity **(d)** like poles attract

7. Which of these is the best example of a testable question? 1

(a) Which sport is exciting?
(b) Is a mango tastier than a pear?
(c) Does temperature affect how fast sugar dissolves?
(d) Is blue a better colour than red?

8. Which group of features suits an animal living in the polar region? 1

(a) gills, fins, scales **(b)** thick fur, fat layer, white colour
(c) thin skin, long legs, hump **(d)** feathers, hollow bones, beak

9. Which pair correctly matches the deficiency and disease? 1

(a) iodine, anaemia **(b)** iron, scurvy
(c) vitamin A, night blindness **(d)** vitamin C, rickets

10. A compass kept on a table shows north in a wrong direction. The best explanation is: 1

(a) a magnet or iron object nearby **(b)** the compass is in sunlight
(c) the table is made of wood **(d)** the compass needle is too light

11. C.V. Raman's work on the scattering of light began with his curiosity about why the sea looks blue. This shows that: 1

(a) only costly equipment can lead to discoveries
(b) great discoveries need no observation
(c) curiosity about everyday things can lead to great discoveries
(d) experiments are unnecessary in science

12. The roots of a banyan hanging from branches are called: 1

(a) tuberous roots **(b)** tap roots **(c)** prop roots **(d)** fibrous roots

13. **Assertion (A):** Lack of vitamin C causes scurvy. 1
Reason (R): Citrus fruits like lemon and amla are rich in vitamin C.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** An iron nail is attracted by a magnet. 1
Reason (R): Cobalt and nickel are also magnetic materials.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** Scientists repeat an experiment several times before accepting the result. 1
Reason (R): The result of one trial is always enough to be sure.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why are growing children given more protein-rich food? 2

17. Why do we say 'magnets act through non-magnetic materials'? 2

18. Why do scientists use a control group? Give an example. 2

19. A student says 'all animals that fly are birds'. Is it correct? Give an example. 2

20. Why is playing in sunlight good for children's bones? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain why a mixture of cereal and pulse such as dal-rice gives a better meal than rice alone. 3

22. Explain how the magnetic compass helped in exploring new lands. 3

23. Compare the use of a thermometer with feeling with the hand to judge temperature. 3

24. Explain how the hump of a camel benefits it in the desert. 3

OR

Fruits and vegetables lose vitamins when cut and stored for long. Why and what to do? 3

25. How can you show the Earth's magnetism with a simple experiment? 3

OR

Describe two ways in which a scientific idea may be proved wrong. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Aarav's tiffin usually has only chips and biscuits. His teacher in Chennai explained that such packaged junk food has a lot of fat, sugar and salt but little else. She asked the class to read food labels and to eat a variety of foods such as dal, vegetables, fruits, rotis and milk. 5

(i) Which nutrient in dal helps growth?

(ii) Name a nutrient in fruits and vegetables that protects the body from disease.

(iii) What can we learn by reading a food label?

(iv) Why is a mix of foods better than only chips and biscuits?

27. Plan a balanced day's menu for a 12-year-old, justify each meal's nutrients and mention what to avoid. 5

OR

A box has three unlabelled bars: a magnet, an iron bar and a copper bar. Describe a method with only the bars to identify each. 5

END OF QUESTION PAPER — Total: 50 Marks

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